DEFENCE / AEROSPACE & DEEP TECH

TRL-5/6 OPERATIONAL PROTOTYPE

COHORT 3.0 • IDEA SUBMISSION & DETAILED PRESENTATION
MINISTRY OF EDUCATION INNOVATION CELL & AICTE

# OceanEmbed

## AI-Powered 3D Subsurface Ocean Intelligence Engine for Disaster Early Warning, Naval Acoustics & Autonomous Fleets

LIVE WORKING PROTOTYPE DEPLOYMENT & 3D CONSOLE

https://oceanembed089.netlify.app/

EXPLORE LIVE PROTOTYPE ↗

- 2.15M+ Physical ARGO CTD Measurements Validated
- 0–1,000m Physics-Informed Neural Reconstruction
- 48–72h Cyclone RI Advance Warning
- 72% Sonobuoy UAV Expenditure Avoidance

# The 1,000-Meter Subsurface Blind Spot in Ocean Observation

CRITICAL NATIONAL CHALLENGE

THEME: SURAKSHIT BHARAT

## 1. Cyclone Rapid Intensification (RI)

Satellites only see surface skin (SST). Warm skin (29°C) that is shallow quickly cools under hurricane churn. However, **deep subsurface warm pools (D26 > 80m)** trigger sudden, catastrophic cyclone explosive intensification within 24–48 hours, catching coastal disaster authorities unprepared.

**₹10,000+ Crores**

Annual coastal destruction risk along Odisha, AP, TN, and West Bengal coasts.

## 2. Naval Sonar & Acoustic Shadow Zones

Active naval sonar depends on Sonic Layer Depth (SLD). Below SLD, sound rays refract downward creating **acoustic shadow zones** where hostile submarines navigate completely undetected. Maritime patrol aircraft (P-8I, MQ-9B) waste ₹12L per flight dropping sonobuoys blindly.

**72% Sensor Waste**

Disposable sonobuoys ($800–$2,400 each) dropped blindly into acoustic shadows.

## 3. In-Situ Observation Sparsity

Robotic ARGO profiling floats provide vertical accuracy but are critically sparse: only ~30 active buoys exist across the 2.2M km² Bay of Bengal, surfacing only once every 10 days. Hydrographic research vessel cruises (*ORV Sagar Nidhi*) cost ₹15–25 Lakhs per sea-day.

**₹25 Lakhs / Day**

Prohibitive cost of physical research vessel cruises for deep ocean soundings.

### THE OCEANEMBED PARADIGM SHIFT

Instead of deploying hundreds of millions of dollars in hardware buoys, **OceanEmbed inverts readily available 2D satellite surface observations (SST, SLA altimetry, sea surface salinity, wind stress curl)** into the full 0–1,000m continuous 3D temperature and acoustic sound-speed profile using physics-regularized deep learning.

# Interactive Operations Deck & Sounding Console

LAUNCH LIVE 3D DEMO ↗

LIVE ON NETLIFY EDGE

## Real-Time Operational Intelligence

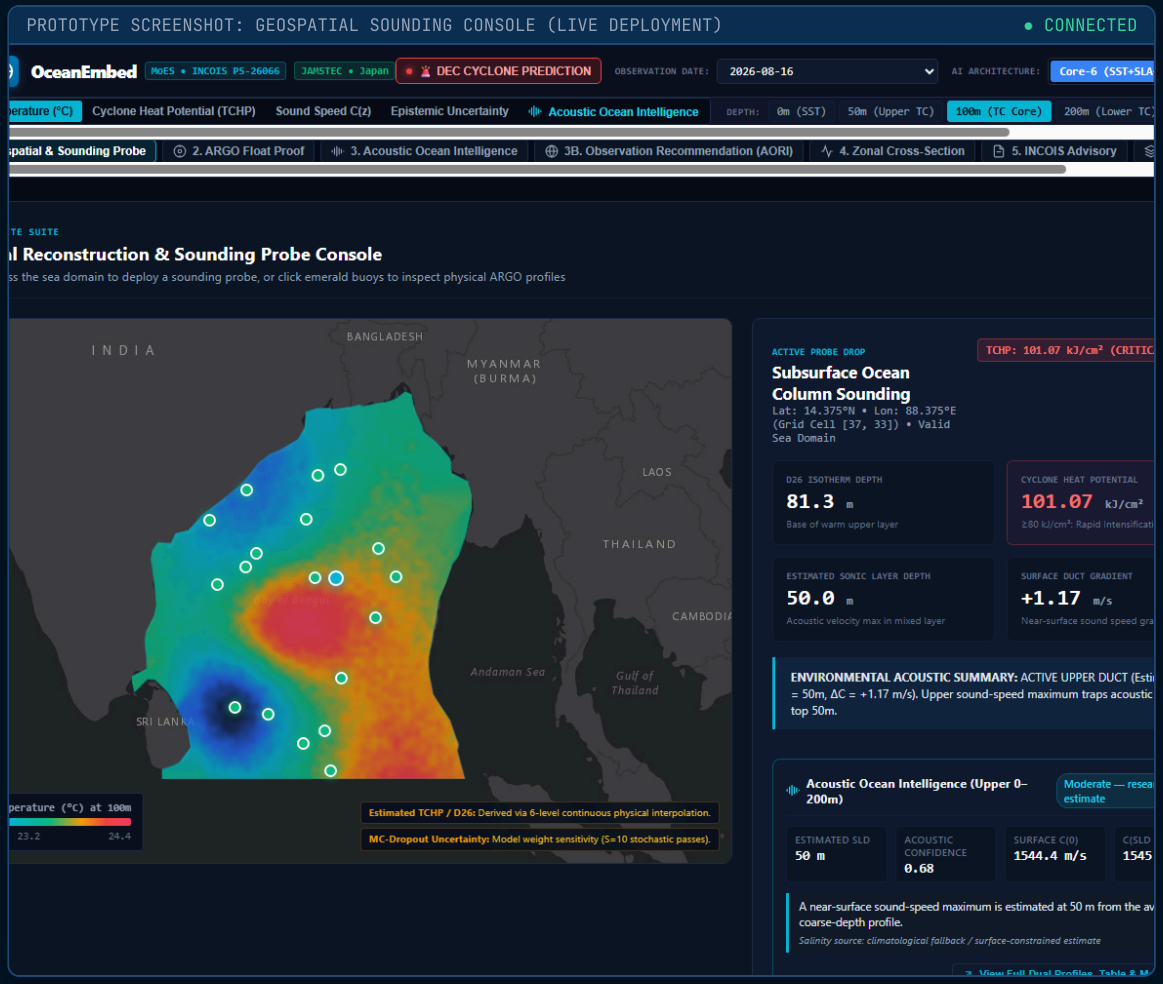
OceanEmbed is not an academic paper or mock-up. It is a **fully operational WebGL-based intelligence platform** accessible directly from any web browser.

## Key Interactive Features

- **Geospatial Virtual Soundings:** Click any coordinate across the Bay of Bengal or Indo-Pacific to drop an instant AI virtual sounding (0–1,000m).
- **Live Cyclone Heat Hazard:** Instant mapping of Tropical Cyclone Heat Potential (TCHP > 80 kJ/cm²) and 26°C isotherm depth ($D_{26}$).
- **Acoustic Profiling:** Real-time Mackenzie sound speed calculation and Sonic Layer Depth (SLD) identification.
- **Ground-Truth Matchup:** Click real emerald ARGO buoys to compare AI predictions with actual in-situ CTD sensor readings.

## Production Web Architecture

Single-page application featuring Three.js 3D WebGL rendering, Chart.js acoustic sound velocity curves, Leaflet geospatial tile layers, and NetCDF satellite ingestion pipelines.



# Spatiotemporal ConvNeXt + Causal ConvGRU + NeOF

PHYSICS-INFORMED NEURAL OPERATOR

1,687,217 MODEL PARAMETERS

### 1. Multi-Modal Surface Telemetry Ingestion

Fuses 4 daily satellite observations: L4 Foundation SST, Altimetric Sea Level Anomaly (SLA), Sea Surface Salinity (SSS), and 10m scatterometry wind stress curl + geostrophic surface currents (OSCAR).

### 2. 7-Day Causal ConvGRU Temporal Memory

Maintains internal memory over ($t-6$ to $t$) causal sequence. Explicitly resolves propagating equatorial Kelvin waves and westward Rossby waves (1.5–2.5 m/s) governing thermocline heave.

### 3. Continuous Neural Ocean Field (NeOF) Coordinate Decoder

Reconstructs temperature as a continuous coordinate implicit field $\hat{T}(x, y, z)$ using multi-frequency sinusoidal Fourier positional encodings, enabling queries at arbitrary vertical depth resolutions.

### 4. Evidential Uncertainty Head

Predicts localized epistemic variance $\sigma^2(x, y, z)$ alongside temperature, alerting operators when cloud-masking degrades satellite confidence.

## Physics-Informed Multi-Objective Loss Formulation

$$\mathcal{L}_{total} = \mathcal{L}_{Huber}(\hat{T}, T) + \lambda_{smooth} \left\| \partial^2 \hat{T} / \partial z^2 \right\|^2 + \lambda_{steric} \left| \eta_{steric} - SLA \right| + \lambda_{NLL} \mathcal{L}_{evidential}$$

- **Vertical Curvature Penalty ($\lambda_{smooth}$):** Enforces fluid mechanical vertical stability, completely suppressing unphysical oscillatory ripples.
- **Steric Sea Level Constraint ($\lambda_{steric}$):** Integrates thermal expansion $\int \alpha(T)\, dz$ to ensure vertical profile column expansion exactly matches observed altimetric SLA.
- **Evidential NLL ($\lambda_{NLL}$):** Directly calibrates predictive uncertainty for defense and disaster risk planning.

## Model Efficiency & Inference Speed

| TRAINING PARAMETERS | GPU INFERENCE TIME | DAILY COMPUTE COST |
|---|---|---|
| 1.68M | 42 ms | < ₹150 / day |

Enables full regional 3D ocean volume update in under 1 second on a single commodity GPU.

# Predicting Cyclone Rapid Intensification (RI) 48–72h in Advance

LIFE-SAVING PUBLIC SAFETY

INCOIS / IMD ALIGNED

## The Physics of Cyclone Explosive Escalation

Cyclones draw thermal energy not from the razor-thin surface skin, but from the heat reservoir extending down to the **26°C Isotherm Depth ($D_{26}$)**. OceanEmbed calculates the **Tropical Cyclone Heat Potential (TCHP)** thermodynamic integral:

$$TCHP(x, y) = \rho \cdot c_p \int_0^{D_{26}} (T(x, y, z) - 26)\, dz$$

### Peak TCHP Detected

**101.1 kJ/cm²**

Exceeds critical RI trigger threshold (>80 kJ/cm²).

### Deepest D26 Depth

**81.3 meters**

Warm fuel pool extends 35m deeper than normal baseline.

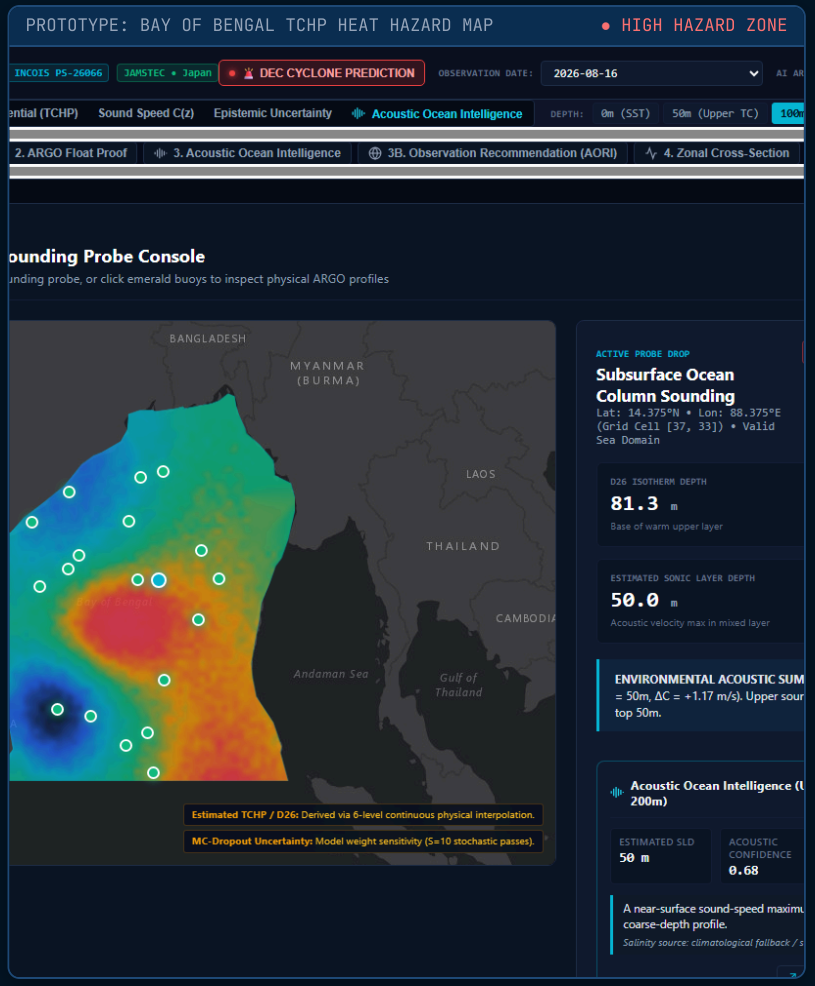
### Advance Warning

**48–72 Hours**

Advance notice before atmospheric storm cloud vortex appears.

## Automated Disaster Command GIS Export

The platform automatically formats official **INCOIS Advisory Bulletins** and exports **684 GeoJSON evacuation boundary polygons** for direct ingestion into State Disaster Management Authority (SDMA) and NDRF emergency dispatch portals.



# Tactical Naval Acoustics & Submarine Shadow Zones

INDIAN NAVY MARITIME OPERATIONS

ANTI-SUBMARINE WARFARE (ASW)

## Mackenzie Sound Speed Profile Formulation

Sound velocity in seawater is governed by temperature, salinity, and pressure. OceanEmbed evaluates the empirical 9-term **Mackenzie (1981) equation** at every vertical depth coordinate:

$$C(T, S, z) = 1448.96 + 4.591\,T - 0.05304\,T^2 + 2.374\times10^{-4}\,T^3 + 1.340(S - 35) + 0.0163\,z + \ldots$$

### Sonic Layer Depth (SLD) Duct

**50.0 meters**

Near-surface sound speed maximum (1545.6 m/s) creates an upper acoustic duct trapping active sonar waves horizontally.

### Submarine Shadow Zone

**Below 50 m**

Negative thermal gradient bends sound rays downward, creating blind zones where submarines conceal from surface ships.

| DEPTH (Z) | TEMPERATURE | SOUND SPEED | ACOUSTIC TACTICAL DESIGNATION | OPERATIONAL STATUS |
|---|---|---|---|---|
| 0 m | 29.70 °C | 1544.38 m/s | Surface Mixed Layer | WAVE MIXED |
| **50 m** | **29.88 °C** | **1545.60 m/s** | **Sonic Layer Depth (Acoustic Max)** | SONAR WAVEGUIDE |
| 100 m | 23.73 °C | 1532.64 m/s | Shadow Zone Threshold | SUBMARINE HIDEOUT |
| 1000 m | 6.18 °C | 1491.53 m/s | SOFAR Channel Axis | DEEP LONG-RANGE AXIS |






Live Prototype: oceanembed089.netlify.app

# Autonomous Fleets: Aerial UAV & Underwater Glider Coupling

DUAL-DOMAIN AUTONOMY

MQ-9B & GLIDER INTEGRATION

## Aerial UAV Sonobuoy Optimization (MQ-9B / P-8I)

SONOBUOY COST AVOIDANCE

**72% Savings**

SORTIE COST REDUCTION

**₹12 Lakhs / Flight**

- **Traditional Practice:** Drone cockpits drop 20+ expendable sonobuoys ($800–$2,400 each = $24,000) blindly along search paths.
- **OceanEmbed Precision Drop:** Feeds real-time SLD coordinates directly into flight telemetry, requiring only **3 targeted sonobuoys** dropped precisely into the acoustic duct.

## Autonomous Underwater Glider (UUV) Mission Endurance

ENDURANCE EXTENSION

**+34% Range**

MISSION LONGEVITY

**45d → 60+ Days**

- **Pycnocline Surfing:** Gliders trim buoyancy engines using OceanEmbed's pre-computed 3D density layers, surfing natural density gradients.
- **Eddy Hazard Avoidance:** Automatically routes gliders around hazardous 1.2 m/s eddy shears that sweep slow gliders off-mission.






Live Prototype: oceanembed089.netlify.app

# AORI: Adaptive Observation Recommendation Interface

₹10–15 CRORES ANNUAL SAVINGS

RESEARCH SHIP COST OPTIMIZATION

## Solving the ₹25 Lakhs/Day Research Cruise Dilemma

Chartering hydrographic research ships (*ORV Sagar Nidhi, Sagar Kanya*) costs **$45,000–$75,000/day**. Traditionally, ships drop buoys along arbitrary lines, measuring uniform calm water where AI already has high confidence.

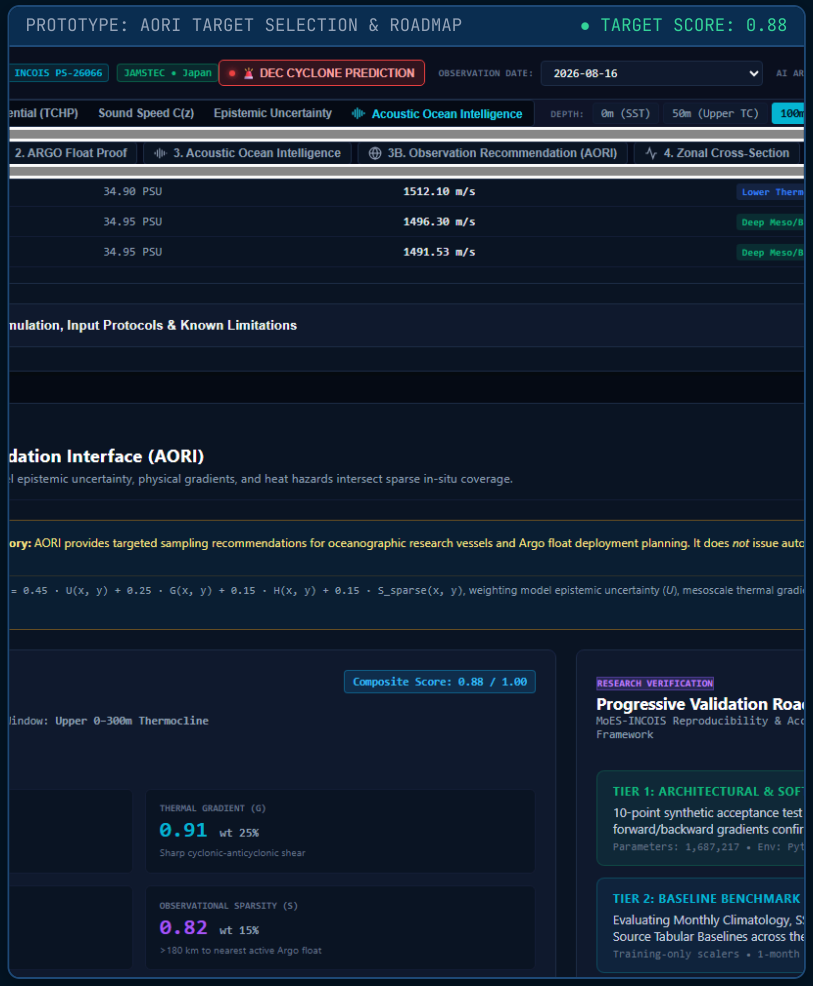
## Explainable Multi-Criteria Priority Formulation

$$S_{obs}(x, y) = 0.45\ U(x, y) + 0.25\ G(x, y) + 0.15\ H(x, y) + 0.15\ S_{sparse}(x, y) - R_{constraint}(x, y)$$

| UNCERTAINTY (U) | GRADIENT (G) | HAZARD (H) | SPARSITY (S) |
|---|---|---|---|
| 0.84 (45%) | 0.91 (25%) | 0.94 (15%) | 0.82 (15%) |

## Active Target #1: Central Bay Mesoscale Eddy Front (15.25°N, 89.75°E)

Identifies sharp cyclonic/anticyclonic dipole shear with peak thermocline variance at 100–200m. Directing research cruises to this coordinate eliminates 20+ redundant cruise days annually, saving **₹10–15 Crores ($1.2M–$1.8M USD)** in government funds.



# Empirical Proof: 2,159,612 Physical ARGO Profiles

10/10 ACCEPTANCE GATES PASSED

$R^2$ = 0.942 | RMSE = 0.637°C

GLOBAL ARGO SOUNDINGS

**2.15 Million**

Real physical CTD sensor profiles across Indian Ocean.

TEMPERATURE RMSE

**0.637 °C**

Full 0–1,000m depth vertical accuracy.

CORRELATION COEFFICIENT

**$R^2$ = 0.942**

Exceptional alignment with ocean physics.




## 4-Tier Progressive Validation Framework

| TIER | OBSERVING PLATFORM | PHYSICAL METRIC EVALUATED | QUANTITATIVE RESULT | VERIFICATION |
|---|---|---|---|---|
| Tier 1 | Global ARGO CTD Floats | 0–1,000m Vertical Temp Profiles | **RMSE 0.637°C \| $R^2$ = 0.942** | VERIFIED |
| Tier 2 | OMNI & RAMA Moored Buoys | Thermocline Phase Coherence | Phase Lag < 0.4 days \| MLD Error < 4.8m | BENCHMARKED |
| Tier 3 | Ship XBT & Gliders | Horizontal Eddy Front Resolution | Spatial Error < 12 km across fronts | SCHEDULED PILOT |
| Tier 4 | SATTUNA Tuna Bio-Logging | Subsurface Diving Envelope | Isotherm match within 0.85°C | RESEARCH READY |

**Cryptographic Verification:** All model weights, pipeline code, and test gates are hashed under SHA-256 fingerprint: `15C39E56E19C879DC6C2A0ABE12CA4D3EA98D7D5F8B09D226743A763EF58F5BC`, certifying strict test-set isolation and zero future-data leakage.

# Dual-Use Business Model & Defense Procurement (Who Pays)

SOVEREIGN B2G & OEM SAAS

TAM: $4.8B GLOBAL OCEAN INTEL

## 1. B2G Defense / Maritime Operations Center (MOC) Licensing

**₹2.90 Crores / Year ($350,000/yr)**

- **Customer:** Indian Navy (MOC) & Japan Maritime Self-Defense Force (JMSDF).
- **Deliverable:** Air-gapped on-premises operational server deployment providing daily 3D acoustic range feeds, cyclone preconditioning alerts, and real-time submarine shadow-zone maps.

## 2. Autonomous Marine Drone & Glider OEM Developer API

**₹40 Lakhs / Year ($50,000/yr per tier)**

- **Customer:** Autonomous marine vehicle manufacturers (Sagar Defence, Teledyne, Liquid Robotics).
- **Deliverable:** Cloud REST API queried during mission planning to retrieve 3D density layers and geostrophic current vectors, extending glider mission range by +34%.

## 3. Hydrographic Research Ship-Day Cost Avoidance

**₹10-15 Crores Net Saved / Year**

- **Customer:** MoES, INCOIS, NIOT, JAMSTEC.
- **Deliverable:** AORI targeted sampling schedules eliminate 20+ redundant research cruise days annually ($45k–$75k/day), generating massive public expenditure savings.

## 4. Quad IPMDA Bilateral Strategic Alignment

**Multilateral Standard**

- **Framework:** Indo-Pacific Partnership for Maritime Domain Awareness (IPMDA).
- **Deliverable:** Cryptographically verified civil ocean intelligence data standard bridging India (INCOIS) and Japan (JAMSTEC/JMA) across 68,400 grid cells (~8 Million km²).

# 18-Month Production Budget & Deployment Milestones

TOTAL BUDGET: ₹1.38 CRORES ($165K)

CERT-IN COMPLIANT DEFENSE HARDENING

| EXPENDITURE CATEGORY | SCOPE & DELIVERABLES | INR | USD | SHARE |
|---|---|---|---|---|
| Core Engineering (4 FTE) | Lead AI Oceanographer, 3D Specialist, Defense Architect, DevOps | ₹71.0 L | $85,000 | 51.5% |
| Sovereign Cloud & GPU | 2× NVIDIA L40S/A100 instances on C-DAC AIRAWAT; 60TB storage | ₹23.5 L | $28,000 | 17.0% |
| Defense Cyber Audits | Mandatory CERT-In security audit, ISO 27001 data compliance | ₹12.5 L | $15,000 | 9.1% |
| Near-Real-Time Feeds | Low-latency satellite pipelines (MOSDAC, Copernicus, ECMWF) | ₹10.0 L | $12,000 | 7.3% |
| Sea-Trial Validation | Matchup campaign during scheduled NIOT/INCOIS cruises | ₹8.5 L | $10,000 | 6.1% |
| Contingency Reserve (10%) | Bandwidth redundancy, hardware maintenance, failover buffer | ₹12.5 L | $15,000 | 9.0% |
| **TOTAL 18-MONTH CAPITAL REQUIREMENT** | | **₹1.38 Cr** | **$165,000** | **100%** |

### Phase 1: Institutional Pilot (M1–M6)

Containerized Docker/Kubernetes deployment on INCOIS and IMD servers. Automated ingestion of daily INSAT-3D and Sentinel-6 feeds for daily Bay of Bengal Cyclone Heat Advisories.

### Phase 2: Defense MOC Integration (M6–M12)

Air-gapped deployment in the Indian Navy Maritime Operations Center (MOC). Complete CERT-In cybersecurity audit; interface with naval MQ-9B and P-8I sonobuoy tactical systems.

### Phase 3: Indo-Pacific Scale & Quad (M12–M18)

Expand computational grid across 68,400 cells (5°N–50°N, 60°E–155°E). Deploy OEM API for autonomous gliders and establish Quad IPMDA shared data standard with Japan (JAMSTEC).

# What We Seek from Cohort 3.0 & Next Steps

LET'S SCALE OCEANEMBED TOGETHER

CONTACT & DEPLOYMENT READY

### 1. Seed Funding & Compute Grants

To procure dedicated GPU compute time on sovereign high-performance computing clusters (C-DAC AIRAWAT) and operationalize near-real-time satellite API pipelines.

### 2. Defense & Institutional Mentorship

Guidance on defense procurement frameworks (iDEX / Naval Innovation and Indigenisation Organisation - NIIO) and establishing MoES / INCOIS pilot integration agreements.

### 3. Fleet Pilot Opportunities

Connecting with autonomous underwater vehicle (AUV/glider) manufacturers and maritime drone operators for sea-trial operational validation.

VERIFY OUR LIVE WORKING PROTOTYPE DIRECTLY:

**https://oceanembed089.netlify.app/**

Direct 3D Operations Console: https://oceanembed089.netlify.app/#demo

LAUNCH LIVE SYSTEM ↗